Section 2.1 Angles, similarity & parallel lines

角度、相似与平行线

Method 1 / 方法 1 [GE01-M01]

Similarity is a statement about ordered correspondence, not a picture's appearance. Prove two angles, name matching vertices and choose a directed scale before transferring lengths, areas or volumes. Parallel cuts and reflections often create the needed triangles.

相似是有序对应关系，不是图形看起来相似。先证明两角、写明对应顶点并选择有向比例，再转移长度、面积或体积；平行截线与反射常能产生所需三角形。

Parallel cuts turn heights into linear scales and areas into squares

平行截线将高度转为线性比、面积转为平方比

Procedure

1. Locate the shared apex and prove the cut triangle is similar by parallel-line angles.

2. Its linear scale equals the ratio of heights measured from that apex.

3. Use the square of scale for area and cube for similar-solid volume; subtract complements only afterward.

4. For equal-area bands, cumulative area determines a square-root length scale, not equal length spacing.

通用步骤

1. 找到公共顶点，由平行线角证明截出三角形相似。

2. 线性相似比等于从该顶点量起的高之比。

3. 面积用比例平方，相似立体体积用立方；之后再扣补集。

4. 等面积条带应由累积面积求平方根长度比例，不是等距截长。

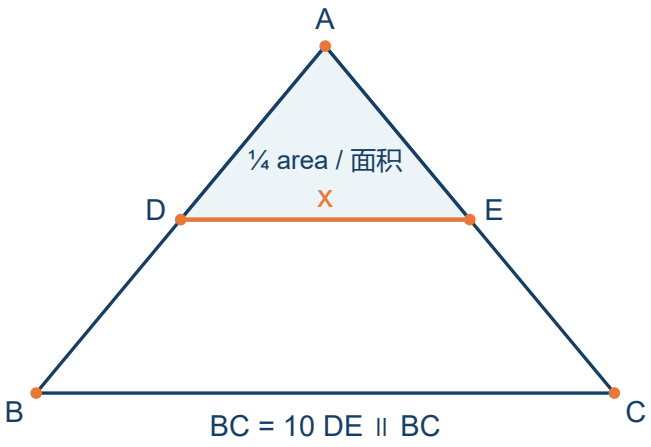
Section 2.1 Angles, similarity & parallel lines

角度、相似与平行线

Method 1 / 方法 1 [GE01-M01] (continued / 续)

> Example. A triangle has base ten. A parallel chord cuts off one quarter of its area at the apex. Find the chord length.

> 例题：三角形底边十，一条平行截线在顶端截出原面积的四分之一，求截段长。



Worked solution

1. The small and original triangles are similar, with positive linear scale t.

2. Their area ratio gives $t^2 = 1/4$, so $t = 1/2$.

3. The chord is $10t = 5$. The remaining trapezoid has three quarters of the original area.

例题解析

1. 小三角形与原者相似，正线性比例设为 t。

2. 由面积比 $t^2 = 1/4$ 得 $t = 1/2$。

3. 截段为 $10t = 5$，余梯形面积占原来的四分之三。

Section 2.1 Angles, similarity & parallel lines

角度、相似与平行线

Method 1 / 方法 1 [GE01-M01] (continued / 续)

Equal-height strips usually have unequal areas.

等高条带通常并不等面积。

Extend trapezoid legs to their common apex. Equal area on the two sides of an intermediate parallel section of length x gives $x^2 = (a^2 + b^2)/2$, where a, b are the two base lengths.

A half-turn about a triangle centroid leaves three exterior corner triangles of linear scale $1/3$, so the union area is $4/3$ of the original.

For a rectangle of width w inscribed above a triangle base of length b and altitude h, similarity gives rectangle height $h(1 - w/b)$.

延长梯形两腰至公共顶点。中间平行截段长 x 将梯形等面积分开时，若底长为 a, b，则 $x^2 = (a^2 + b^2)/2$。

三角形绕重心半周旋转，产生三个线性比例 $1/3$ 的外角三角形，故并集面积是原来的 $4/3$。

三角形底长 b、高 h，内部贴底矩形宽为 w 时，相似给出矩形高 $h(1 - w/b)$。

Practice / 配套练习: AIME-2000-I-Q08 (WB p. 444), AIME-2000-II-Q06 (WB p. 246), AIME-2001-I-Q07 (WB p. 224), AIME-2001-II-Q07 (WB p. 313), AIME-2003-II-Q06 (WB p. 248), AIME-2004-I-Q11 (WB p. 453), AIME-2006-I-Q07 (WB p. 249), AIME-2008-I-Q02 (WB p. 241), AIME-2010-I-Q13 (WB p. 255), AIME-2011-I-Q08 (WB p. 251), AIME-2012-I-Q08 (WB p. 447), AIME-2015-II-Q07 (WB p. 226), AIME-2018-II-Q07 (WB p. 227)

Section 2.1 Angles, similarity & parallel lines

角度、相似与平行线

Method 2 / 方法 2 [GE01-M02]

Ordered similar triangles transfer ratios and secant products

有序相似三角形转移比例与割线乘积

Procedure

1. Write a named similarity in matching vertex order, justified by two angles.

2. List corresponding sides before choosing a ratio; crossed diagrams can reverse the visual order.

3. Multiply equal ratios to produce a length product when an intermediate side is unknown.

4. For cyclic or tangent configurations, use directed angles first, then verify the positive ray order.

通用步骤

1. 以对应顶点顺序写出相似，并用两个角证明。

2. 选比例前先列对应边；交叉图形可能颠倒视觉次序。

3. 中间边未知时，可相乘比例得到长度乘积。

4. 圆或切线构型先用有向角，再核实正射线点序。

> Example. Triangles ABC, DEF are similar in that order. If $(AB, BC, CA) = (6, 9, 12)$ and $DE = 4$, find EF, FD and their area ratio.

> 例题：ABC, DEF 按所写顺序相似，$(AB, BC, CA) = (6, 9, 12)$ 且 $DE = 4$，求 EF, FD 及面积比。

Section 2.1 Angles, similarity & parallel lines

角度、相似与平行线

Method 2 / 方法 2 [GE01-M02] (continued / 续)

Worked solution

1. The correspondence is $AB \leftrightarrow DE$, $BC \leftrightarrow EF$, $CA \leftrightarrow FD$.

2. The scale from the first to second is $4/6 = 2/3$, giving $EF = 6, FD = 8$.

3. The second-to-first area ratio is $(2/3)^2 = 4/9$. All three side ratios agree.

例题解析

1. 对应关系为 $AB \leftrightarrow DE, BC \leftrightarrow EF, CA \leftrightarrow FD$。

2. 从前者到后者的比例为 $4/6 = 2/3$，故 $EF = 6, FD = 8$。

3. 后者对前者面积比为 $4/9$，三组边比均一致。

Writing only “the triangles are similar” can hide the crucial correspondence.

只写三角形相似，可能掩盖关键对应关系。

For a right-triangle altitude foot D on hypotenuse AB, similarity gives $AD \cdot AB = AC^2$, $BD \cdot AB = BC^2$ and $CD^2 = AD \cdot BD$.

For an infinite geometric area construction, first identify the retained linear scale and square it; target pieces may require subtracting two similar areas before summing.

直角三角形高的垂足 D 在斜边 AB 上时，相似给出 $AD \cdot AB = AC^2, BD \cdot AB = BC^2, CD^2 = AD \cdot BD$。

Section 2.1 Angles, similarity & parallel lines

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Method 2 / 方法 2 [GE01-M02] (continued / 续)

无穷等比面积构造应先确定保留的线性比例并平方；每个目标块可能需先作两相似面积之差，再求和。

Practice / 配套练习: AIME-2001-II-Q13 (WB p. 230), AIME-2002-II-Q14 (WB p. 340), AIME-2006-II-Q12 (WB p. 233), AIME-2007-I-Q12 (WB p. 253), AIME-2007-I-Q15 (WB p. 234), AIME-2013-I-Q13 (WB p. 235), AIME-2016-I-Q06 (WB p. 316), AIME-2016-I-Q15 (WB p. 299), AIME-2016-II-Q10 (WB p. 287), AIME-2019-I-Q06 (WB p. 228), AIME-2019-II-Q15 (WB p. 304), AIME-2023-II-Q03 (WB p. 266), AIME-2015-I-Q06 (WB p. 286)

Section 2.1 Angles, similarity & parallel lines

角度、相似与平行线

Method 3 / 方法 3 [GE01-M03]

Extend parallel sides to reveal a homothety

延长平行边发现位似

Procedure

1. Extend the relevant sides until triangles share an apex or an auxiliary parallelogram appears.

2. Use parallel sides to identify corresponding angles and the homothety center.

3. Keep the scale signed when the image lies across the center; length and area use its absolute value and square.

4. For three parallel-to-side lines, affine or barycentric level equations give the signed scale directly.

通用步骤

1. 延长相关边，使三角形共享顶点或出现辅助平行四边形。

2. 由平行边确定对应角及位似中心。

3. 像在中心另一侧时保留负位似比，长度用绝对值、面积用平方。

4. 三条分别平行于三边的直线，可用仿射或重心坐标水平式直接求有向比例。

> Example. In triangle ABC, choose $D \in AB, E \in AC$ with $AD/AB = AE/AC = 2/5$. Reflect D, E in point A to $D\prime, E\prime$. Find the signed homothety scale from ABC to $AD\prime E\prime$ and their area ratio.

Section 2.1 Angles, similarity & parallel lines

角度、相似与平行线

Method 3 / 方法 3 [GE01-M03] (continued / 续)

> 例题：三角形中，$D \in AB, E \in AC$ 满足 $AD/AB = AE/AC = 2/5$。将 D, E 关于点 A 作中心反射得 $D\prime, E\prime$。求从 ABC 到 $AD\prime E\prime$ 的有向位似比及面积比。

Worked solution

1. The equal fractions and shared angle give $ADE \sim ABC$, with $DE \parallel BC$ and scale $2/5$.

2. With A as vector origin, $D\prime = -2B/5$ and $E\prime = -2C/5$. Thus the signed homothety scale is $-2/5$: both images lie on the opposite rays.

3. Ordinary area uses the square, so $[AD\prime E\prime]/[ABC] = 4/25$. The negative length scale does not make area negative.

例题解析

1. 两边比例相等且夹角共用，得 $ADE \sim ABC$，且 $DE \parallel BC$，线性比例为 $2/5$。

2. 以 A 为向量原点，则 $D\prime = -2B/5, E\prime = -2C/5$，有向位似比为 $-2/5$，两个像点均在反向射线上。

3. 普通面积取比例平方，故 $[AD\prime E\prime]/[ABC] = 4/25$；负线性比例不会产生负的普通面积。

Parallel-sided triangles may be oppositely oriented.

各边平行的两个三角形可能方向相反。

For barycentric coordinates $\alpha + \beta + \gamma = 1$, lines $\alpha = a, \beta = b, \gamma = c$ form a triangle similar to the reference triangle with absolute ratio $|1 - a - b - c|$.

Section 2.1 Angles, similarity & parallel lines

角度、相似与平行线

Method 3 / 方法 3 [GE01-M03] (continued / 续)

Common external tangents of unequal circles meet at their external homothety center; center distances have the radius ratio.

If three parallel corner cuts leave a six-equal-sided hexagon inside a triangle of sides a, b, c, its common side s satisfies $1 = s(1/a + 1/b + 1/c)$. On the side of length a, the two removed lengths are as/b and as/c, leaving the middle length s; divide their sum by a.

在重心坐标 $\alpha + \beta + \gamma = 1$ 中，三线 $\alpha = a, \beta = b, \gamma = c$ 所围三角形对参考三角形的绝对相似比为 $|1 - a - b - c|$。

两不等圆公外切线交于外位似中心，到两圆心距离比等于半径比。

若三条平行截角线在边长为 a, b, c 的三角形内留下六边等长的六边形，其公共边长 s 满足 $1 = s(1/a + 1/b + 1/c)$。沿长为 a 的边，两端切去段为 $as/b, as/c$，中央保留 s；将三段之和除以 a 即得。

Practice / 配套练习: AIME-2002-II-Q13 (WB p. 231), AIME-2004-II-Q13 (WB p. 232), AIME-2008-II-Q05 (WB p. 220), AIME-2009-I-Q05 (WB p. 221), AIME-2019-II-Q07 (WB p. 229), AIME-2022-II-Q07 (WB p. 337), AIME-2024-II-Q05 (WB p. 223)

Section 2.1 Angles, similarity & parallel lines

角度、相似与平行线

Method 4 / 方法 4 [GE01-M04]

Equal lengths and polygon angle sums determine the orientation

等长与多边形角和确定方向

Procedure

1. Mark equal sides and assign the matching base angles in every isosceles triangle.

2. Use the polygon interior sum or exterior turns to determine the remaining angles.

3. An obtuse angle in an isosceles triangle must be its apex; use this to decide which sides are equal.

4. Check positive side portions and convexity before accepting a trigonometric candidate.

通用步骤

1. 标记等边，并给各等腰三角形分配相等底角。

2. 利用多边形内角和或外转角求余角。

3. 等腰三角形中的钝角必为顶角，可据此决定相等的两边。

4. 接受三角方程候选前，检查分段正性及凸性。

> Example. An isosceles triangle has one angle 120°. Determine all its angles and the ratio of its base to an equal side.

> 例题：等腰三角形有一角为 120°，求三角及底边与腰的比。

Worked solution

1. The 120° angle cannot be a base angle, since two such angles exceed 180°.

Section 2.1 Angles, similarity & parallel lines

角度、相似与平行线

Method 4 / 方法 4 [GE01-M04] (continued / 续)

2. The other two angles are $(180-120)/2=30^\circ$.

3. For equal side a, cosine law gives base squared $2a^2-2a^2\cos 120^\circ=3a^2$, so the ratio is $\sqrt{3}$.

例题解析

1. 120° 不可能是底角，否则两个这样的角超过 180°。

2. 另两角均为 30°。

3. 腰长为 a 时，余弦定理给出底边平方 $3a^2$，故比例为 $\sqrt{3}$。

Do not assign the two equal sides from a sketch alone.

不能仅凭草图决定哪两边相等。

Practice / 配套练习: AIME-2000-I-Q14 (WB p. 272), AIME-2006-II-Q01 (WB p. 240), AIME-2007-II-Q03 (WB p. 259), AIME-2011-II-Q03 (WB p. 131), AIME-2013-I-Q12 (WB p. 398), AIME-2021-II-Q02 (WB p. 244), AIME-2025-II-Q05 (WB p. 282)

Section 2.1 Angles, similarity & parallel lines

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Method 5 / 方法 5 [GE01-M05]

Bisector reflections expose midpoint lines

角平分线反射显露中点线

Procedure

1. Reflect a vertex across a known internal angle bisector to place it on the other side ray at the same distance.

2. Combine two such reflections with equal-distance or midpoint information.

3. Use the midpoint theorem or an isosceles median to convert the reflected geometry into a parallel segment.

通用步骤

1. 关于已知内角平分线反射顶点，使其落在另一边射线上且到顶点距离不变。

2. 把两次反射与等距或中点信息结合。

3. 用中点定理或等腰中线，将反射构型化为平行线段。

> Example. In triangle ABC, $AB = AC$ and D is the midpoint of BC. Prove AD is both an angle bisector and perpendicular to BC.

> 例题：三角形中 $AB = AC$，D 为 BC 中点，证明 AD 既平分角又垂直于 BC。

Worked solution

1. Triangles ABD, ACD have equal sides $AB = AC$, $BD = DC$ and common AD.

2. SSS congruence gives equal angles at A and equal adjacent angles ADB, ADC.

Section 2.1 Angles, similarity & parallel lines

角度、相似与平行线

Method 5 / 方法 5 [GE01-M05] (continued / 续)

3. The latter sum to 180°, so each is 90°; hence both conclusions hold.

例题解析

1. ABD, ACD 有 $AB = AC, BD = DC$ 及公共边 AD。

2. SSS 全等给出 A 处两角相等，以及相邻的 ADB, ADC 相等。

3. 后两角和为 180°，故各为 90°，两结论均成立。

An angle bisector becomes a median only after an isosceles relation is proved.

只有证明等腰后，角平分线才能直接视为中线。

Practice / 配套练习: AIME-2011-I-Q04 (WB p. 310), AIME-2022-I-Q03 (WB p. 312)

Section 2.1 Angles, similarity & parallel lines

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Method 6 / 方法 6 [GE01-M06]

Similar pieces must fit the entire original shapes

相似切块必须容纳于整个原图形

Procedure

1. List all possible side correspondences and cut orientations compatible with the required angles.

2. Express every cut length or altitude in a common linear scale.

3. Enforce the full original side lengths and pairwise non-overlap inequalities, rather than one visible constraint.

4. Construct the equality case to show the best candidate is attainable.

通用步骤

1. 列出符合角条件的全部边对应及切割方向。

2. 把每条截长或高写为共同线性比例。

3. 施加原图全部边长及两两不重叠不等式，不能只检查一个明显限制。

4. 构造等号情形，证明最优候选可达。

> Example. A 3–4–5 right triangle is placed axis-aligned in a 4×5 rectangle. What is the largest scale if its legs must lie along rectangle sides?

> 例题：三、四、五直角三角形两直角边沿长方形边方向放入四乘五矩形，求最大线性比例。

Section 2.1 Angles, similarity & parallel lines

角度、相似与平行线

Method 6 / 方法 6 [GE01-M06] (continued / 续)

Worked solution

1. With the three-leg along the four-side and four-leg along the five-side, $t \leq \min(4/3, 5/4) = 5/4$.

2. The exchanged orientation requires $t \leq \min(4/4, 5/3) = 1$.

3. Thus the maximum is $5/4$, attained by legs $15/4$ and five inside the rectangle.

例题解析

1. 三长边沿四边、四长边沿五边时，$t \leq \min(4/3, 5/4) = 5/4$。

2. 交换方向时，$t \leq \min(4/4, 5/3) = 1$。

3. 故最大比例为 $5/4$，直角边 $15/4$ 与五确能放入矩形。

A local angle match does not guarantee global containment.

局部角度匹配并不保证整体容纳。

Practice / 配套练习: AIME-2004-I-Q09 (WB p. 225), AIME-2011-I-Q08 (WB p. 251), AIME-2013-I-Q12 (WB p. 398)

Section 2.1 Angles, similarity & parallel lines

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Method 7 / 方法 7 [GE01-M07]

Iterated right-triangle altitudes have a constant similarity ratio

迭代直角三角形高具有固定相似比

Procedure

1. Identify the fixed acute angle shared by every new right triangle.

2. Use AA similarity to find the linear ratio from one altitude to the next.

3. Compute the first altitude from leg product divided by hypotenuse.

4. Sum the resulting geometric progression, checking the ratio lies strictly between zero and one.

通用步骤

1. 找出各新直角三角形共享的固定锐角。

2. 由 AA 相似求相继高的线性比例。

3. 第一条高由两直角边乘积除以斜边求得。

4. 求所得等比级数和，并检验公比严格介于零与一。

> Example. Let ABC_0 be right at C_0, with $BC_0 = 3, AC_0 = 4$. Let C_1 be the foot from C_0 to AB; for $n \geq 2$, let C_n be the foot from C_{n-1} to line $C_{n-2}B$. Find $\sum_{n\geq 1} C_{n-1}C_n$.

> 例题：直角三角形 ABC_0 在 C_0 为直角，$BC_0 = 3, AC_0 = 4$。C_1 为 C_0 到 AB 的垂足；对 $n \geq 2$，C_n 为 C_{n-1} 到直线 $C_{n-2}B$ 的垂足。求 $\sum_{n\geq 1} C_{n-1}C_n$。

Section 2.1 Angles, similarity & parallel lines

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Method 7 / 方法 7 [GE01-M07] (continued / 续)

Worked solution

1. The first altitude is $3 \cdot 4/5 = 12/5$.

2. The shared-angle similarity ratio is $3/5$, so subsequent altitudes are multiplied by $3/5$.

3. The total is $(12/5)/(1 - 3/5) = 6$, and convergence follows from $3/5 < 1$.

例题解析

1. 第一高为 $3 \cdot 4/5 = 12/5$。

2. 共享角相似比为 $3/5$，以后每条高乘此比例。

3. 总和为 $(12/5)/(1 - 3/5) = 6$，且因公比小于一而收敛。

The first altitude and the ratio come from different parts of the original triangle.

第一高与公比来自原三角形的不同度量关系。

Practice / 配套练习: AIME-2016-II-Q05 (WB p. 222)